1. Concepts of State, State Variables and State Model

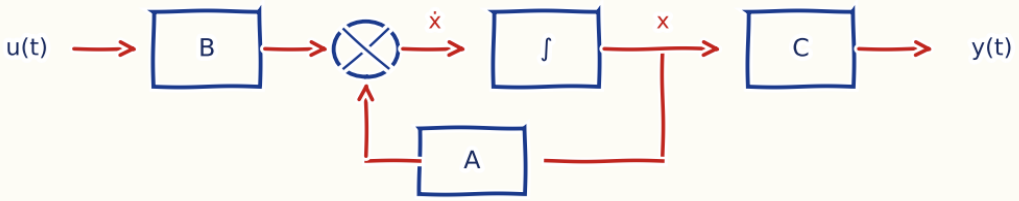
Classical (transfer function) methods deal with single-input single-output LTI systems with zero initial conditions. ***State space analysis*** *(modern control) is a time-domain method that handles MIMO, time-varying, non-linear systems and non-zero initial conditions, and is convenient for computers.*

Term	Definition
State	Minimum set of variables whose values at $t = t_0$, together with inputs for $t \geq t_0$, determine the behaviour of the system for all $t \geq t_0$
State variables	The variables $x_1(t), \ldots, x_n(t)$ that make up the state; n = order of the system (number of energy-storing elements / integrators)
State vector	Column vector $x(t) = [x_1 \ldots x_n]^T$
State space	n-dimensional space whose axes are the state variables; the system trajectory lies in it
State equation	First-order differential equations for the derivatives of the states
Output equation	Algebraic equation giving outputs in terms of states and inputs

State equation: $x'(t) = A\,x(t) + B\,u(t)$

Output equation: $y(t) = C\,x(t) + D\,u(t)$

- $x : n \times 1$ state vector; $u : r \times 1$ input vector; $y : m \times 1$ output vector.
- $A : n \times n$ system (state) matrix; $B : n \times r$ input matrix; $C : m \times n$ output matrix; $D : m \times r$ direct-transmission (feed-forward) matrix (usually zero).



Advantages of state space over transfer function

- Applies to MIMO, non-linear and time-varying systems; initial conditions included.
- Gives internal behaviour (all states), not only input–output; suitable for optimal control, digital computers.
- Can handle controllability and observability analysis; design by pole placement.

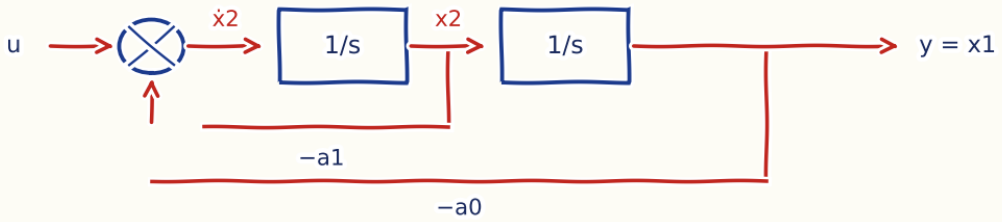
2. Derivation of State Models

(a) From a differential equation / transfer function – phase variables

For $d^n y/dt^n + a_{n-1} d^{n-1} y/dt^{n-1} + \ldots + a_0 y = u$ choose $x_1 = y$, $x_2 = y'$, …, $x_n = y^{(n-1)}$. Then $x'_1 = x_2$, …, $x'_n = -a_0 x_1 - \ldots - a_{n-1} x_n + u$.

$A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-a_0\,-a_1\,-a_2\,]$ (3rd order, rows separated by ;)

$B = [\,0\,;\,0\,;\,1\,]$, $C = [\,1\,0\,0\,]$, $D = 0$ (companion / controllable canonical form)



If the T.F. has numerator, $G(s) = (b_2 s^2 + b_1 s + b_0)/(s^3 + a_2 s^2 + a_1 s + a_0)$, A and B stay the same and **$C = [\,b_0\,b_1\,b_2\,]$**.

Example

$y''' + 6y'' + 11y' + 6y = u$:

$A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-6\,-11\,-6\,]$, $B = [\,0\,;\,0\,;\,1\,]$, $C = [\,1\,0\,0\,]$

(b) From an electrical network

Choose capacitor voltages and inductor currents as state variables. Series R–L–C, input v_i, output v_c; $x_1 = v_c$, $x_2 = i_L$:

$$x'_1 = x_2/C \; ; \; x'_2 = -x_1/L - (R/L)x_2 + v_i/L \; ; \; y = x_1$$
$$A = [\, 0 \;\; 1/C \; ; \; -1/L \;\; -R/L \,] \; , \; B = [\, 0 \; ; \; 1/L \,] \; , \; C = [\, 1 \;\; 0 \,]$$

(c) From a block diagram

- Redraw the diagram so that each block is a simple first-order block $1/(s + a)$ or an integrator $1/s$.
- Assign a state variable to the **output of each integrator / first-order block**.
- Write x' for each block (e.g., block $1/(s+a)$: $x' = -a\,x + \text{input}$) and express the input of each block using states and the reference input.
- Write the output y as a combination of states.

Example: unity-feedback system, $G(s) = 10/[s(s+2)]$ split as $e \rightarrow [10/(s+2)] \rightarrow x_2 \rightarrow [1/s] \rightarrow x_1 = c$, with $e = r - x_1$:

$$x'_2 = -2x_2 + 10(r - x_1) \; , \; x'_1 = x_2$$
$$A = [\, 0 \;\; 1 \; ; \; -10 \;\; -2 \,] \; , \; B = [\, 0 \; ; \; 10 \,] \; , \; C = [\, 1 \;\; 0 \,] \; (\text{check: } |sI - A| = s^2 + 2s + 10 \; \square)$$

(d) Parallel (diagonal) and Jordan forms

Expand $G(s)$ by partial fractions $G(s) = \Sigma\, c_i/(s - p_i)$ for distinct poles. Each term gives $x'_i = p_i x_i + u$; $y = \Sigma\, c_i x_i \rightarrow A = \text{diag}(p_1, \ldots, p_n)$ (decoupled states). For repeated poles the Jordan canonical form has 1's on the superdiagonal.

Transfer function from state model

$$\mathbf{G(s) = Y(s)/U(s) = C\,(sI - A)^{-1}\,B + D}$$

Characteristic equation: $|sI - A| = 0$; eigenvalues of A = poles of the system

Check: $A = [\,0\ 1\,;\,-2\ -3\,]$, $B = [\,0\,;\,1\,]$, $C = [\,1\ 0\,]$: $(sI - A)^{-1} = 1/[(s+1)(s+2)]\,[\,s+3\ 1\,;\,-2\ s\,] \Rightarrow G(s) = 1/[(s+1)(s+2)]$.

3. Diagonalization

Diagonalising A decouples the state equations, making analysis (and e^{At}) easy. Use the transformation $x = M\,z$ where M is the **modal matrix** (columns = eigenvectors):

$$\mathbf{z' = \Lambda\, z + M^{-1}B\,u\,,\ y = C\,M\,z + D\,u\,,\ \Lambda = M^{-1}\,A\,M = diag(\lambda_1, \ldots, \lambda_n)}$$

Steps

- Find eigenvalues λ_i from $|\lambda I - A| = 0$.
- For each λ_i solve $(\lambda_i I - A)\,m_i = 0$ for the eigenvector m_i.
- Form $M = [\,m_1\ m_2 \ldots m_n\,]$ and compute $\Lambda = M^{-1}AM$.

- **Shortcut:** *for a companion (phase-variable) matrix with distinct eigenvalues, M is the Vandermonde matrix: $M = [\,1\ 1\ \dots\ 1\,;\,\lambda_1\ \lambda_2\ \dots\ \lambda_n\,;\,\lambda_1^2\ \lambda_2^2\ \dots\ \lambda_n^2\,;\,\dots\,]$.*

Example

$A = [\,0\ 1\,;\,-2\ -3\,]$: $|\lambda I - A| = \lambda^2 + 3\lambda + 2 = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = -2$.

$$m_1 = [\,1\,;\,-1\,],\ m_2 = [\,1\,;\,-2\,] \Rightarrow M = [\,1\ 1\,;\,-1\ -2\,],$$
$$M^{-1} = [\,2\ 1\,;\,-1\ -1\,]$$
$$\Lambda = M^{-1}AM = [\,-1\ 0\,;\,0\ -2\,]$$

★ *Repeated eigenvalues → generalised eigenvectors → Jordan form $J = M^{-1}AM$. If eigenvalues are complex conjugate use real modal form with 2×2 blocks $[\,\sigma\ \omega\,;\,-\omega\ \sigma\,]$.*

4. Solution of Time-Invariant State Equations

Homogeneous equation ($u = 0$): $x' = Ax$, $x(0) = x_0$

$$x(t) = e^{At}\,x(0) = \Phi(t)\,x(0)$$
$$e^{At} = I + At + A^2t^2/2! + A^3t^3/3! + \dots$$
$$\Phi(t) = L^{-1}\{\,(sI - A)^{-1}\,\}$$

Complete solution (with input)

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

$$y(t) = Cx(t) + Du(t)$$

$$\text{Laplace form: } X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$$

The first term is the ***zero-input (free) response*** *and the second is the* ***zero-state (forced) response*** *= convolution of $\Phi(t)B$ with $u(t)$.*

Methods of computing e^{At}

- ***Laplace transform:*** $\Phi(t) = L^{-1}[(sI - A)^{-1}]$.
- ***Diagonalisation:*** $e^{At} = M\,e^{\Lambda t}\,M^{-1}$ *with* $e^{\Lambda t} = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})$.
- ***Cayley–Hamilton:*** $e^{At} = \alpha_0 I + \alpha_1 A + \ldots + \alpha_{n-1}A^{n-1}$, *where* $e^{\lambda_i t} = \alpha_0 + \alpha_1\lambda_i + \ldots$ *(one equation per eigenvalue).*
- ***Power series*** *(truncated; useful when A is nilpotent).*

Worked example ($A = [\,0\;1\,;\,-2\;-3\,]$)

$(sI - A)^{-1} = 1/[(s+1)(s+2)]\,[\,s+3\;1\,;\,-2\;s\,]$. *By partial fractions:*

$$\Phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

For $B = [\,0\,;\,1\,]$, unit-step input, $x(0) = 0$: $x_1(t) = \int (e^{-(t-\tau)} - e^{-2(t-\tau)})d\tau$ *and* $x_2 = x'_1$:

$$x_1(t) = \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t},\quad x_2(t) = e^{-t} - e^{-2t}$$

5. State Transition Matrix and Its Properties

The **state transition matrix (STM)** $\Phi(t) = e^{At}$ describes how the state propagates from the initial state: $x(t) = \Phi(t - t_0)x(t_0)$ for the zero-input case. It satisfies $\Phi'(t) = A\Phi(t)$ with $\Phi(0) = I$.

#	Property	Meaning
1	$\Phi(0) = I$	Identity at $t = 0$
2	$\Phi^{-1}(t) = \Phi(-t)$	Inverse = time reversal
3	$\Phi(t_1 + t_2) = \Phi(t_1)\Phi(t_2) = \Phi(t_2)\Phi(t_1)$	Semigroup / commutative
4	$[\Phi(t)]^k = \Phi(kt)$	Power property
5	$\Phi(t_2 - t_1)\Phi(t_1 - t_0) = \Phi(t_2 - t_0)$	Transition property
6	$d\Phi/dt = A\Phi(t) = \Phi(t)A$	Derivative; $\Phi'(0) = A$
7	$\Phi(t) = M\, e^{\Lambda t}\, M^{-1}$	Diagonalisation

Proof of (2): $\Phi(t)\Phi(-t) = e^{At}e^{-At} = e^{0} = I \Rightarrow \Phi^{-1}(t) = \Phi(-t)$.

For the example above, check $\Phi(0) = [\ 2-1\ 1-1\ ;\ -2+2\ -1+2\ ] = [\ 1\ 0\ ;\ 0\ 1\ ]$ ☒ and $\Phi'(0) = [\ -2+2\ -1+2\ ;\ 2-4\ 1-4\ ] = [\ 0\ 1\ ;\ -2\ -3\ ] = A$ ☒.

6. Controllability and Observability

Controllability

A system is ***completely state controllable*** *if there exists a control u(t) that can transfer any initial state $x(t_0)$ to any final state in finite time.*

> **Kalman's test: $Q_c = [\, B \; AB \; A^2B \; \dots \; A^{n-1}B \,]$;**
> **controllable iff $\text{rank}(Q_c) = n$ ($|Q_c| \neq 0$ for single input)**

Gilbert's test *(diagonal form, distinct eigenvalues): controllable iff the matrix $M^{-1}B$ has* ***no zero row*** *(a zero row means that mode cannot be affected by the input).*

Observability

A system is ***completely observable*** *if every initial state $x(t_0)$ can be determined from the knowledge of the output y(t) and input over a finite interval.*

> **Kalman's test: $Q_o = [\, C^T \; A^TC^T \; (A^T)^2C^T \; \dots \; (A^T)^{n-1}C^T \,]$;**
> **observable iff $\text{rank}(Q_o) = n$**

Gilbert's test: *in diagonal form, observable iff the matrix CM has* ***no zero column****.*

Examples

(i) $A = [\,0\ 1\,;\,-2\ -3\,]$, $B = [\,0\,;\,1\,]$, $C = [\,1\ 0\,]$: $AB = [\,1\,;\,-3\,] \Rightarrow Q_c = [\,0\ 1\,;\,1\ -3\,]$, $|Q_c| = -1 \neq 0 \rightarrow$ **controllable**. $A^T C^T = [\,0\,;\,1\,] \Rightarrow Q_o = [\,1\ 0\,;\,0\ 1\,]$, rank 2 → **observable**.

(ii) $A = [\,-1\ 0\,;\,0\ -2\,]$, $B = [\,1\,;\,0\,]$: $AB = [\,-1\,;\,0\,]$, $Q_c = [\,1\ -1\,;\,0\ 0\,]$, $|Q_c| = 0 \rightarrow$ **not controllable** (second mode x_2 unaffected by u).

(iii) $A = [\,1\ 0\,;\,0\ -1\,]$, $B = [\,1\,;\,1\,]$, $C = [\,0\ 1\,]$: $Q_c = [\,1\ 1\,;\,1\ -1\,] \rightarrow$ rank 2 (controllable); $Q_o = [\,C^T\ A^T C^T\,] = [\,0\ 0\,;\,1\ -1\,] \rightarrow$ rank 1 → **not observable**.

Point	Controllability	Observability
Question	Can input drive all states?	Can output reveal all states?
Test matrix	$[\,B\ AB\ \ldots\ A^{n-1}B\,]$	$[\,C^T\ A^T C^T\ \ldots\,]$
Gilbert (diagonal)	No zero row in $M^{-1}B$	No zero column in CM
Duality	(A, B) controllable ⇔ (A^T, B^T) observable	(A, C) observable ⇔ (A^T, C^T) controllable

★ **Relation with transfer function: a pole–zero cancellation in G(s) means the system is NOT completely controllable and/or observable. A T.F. gives only the controllable and observable part of the system. Pole placement by state feedback $u = -Kx$ needs controllability.**

7. More Solved Problems (exam type)

Problem 1 – State model from a transfer function with zeros

$G(s) = (s + 3)/(s^3 + 6s^2 + 11s + 6)$. Phase-variable form: $a_0 = 6$, $a_1 = 11$, $a_2 = 6$; numerator $b_0 = 3$, $b_1 = 1$, $b_2 = 0$.

$$\mathbf{A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-6\,-11\,-6\,],\ B = [\,0\,;\,0\,;\,1\,],\ C = [\,3\,1\,0\,],\ D = 0}$$

Poles: $s^3 + 6s^2 + 11s + 6 = (s+1)(s+2)(s+3) \rightarrow -1, -2, -3$ (all in LHP → stable). The numerator zero at -3 cancels the pole at -3, so the system is controllable but **not observable** in this realisation (or vice versa in another).

Problem 2 – Diagonalisation of a 3rd-order system

For the A above, eigenvalues are $-1, -2, -3$ (distinct) and A is in companion form, so the modal matrix is the Vandermonde matrix:

$$\mathbf{M = [\,1\,1\,1\,;\,-1\,-2\,-3\,;\,1\,4\,9\,] \Rightarrow \Lambda = M^{-1}AM = diag(-1, -2, -3)}$$

In the new coordinates $z = M^{-1}x$ each mode evolves independently: $z_i(t) = e^{\lambda_i t} z_i(0) + \int e^{\lambda_i (t-\tau)} (M^{-1}B)_i \, u \, d\tau$.

Problem 3 – e^{At} by Cayley–Hamilton

$A = [\,0\ 1\,;\,-2\ -3\,]$, eigenvalues $-1, -2$. For $n = 2$: $e^{At} = \alpha_0 I + \alpha_1 A$ with

$$e^{-t} = \alpha_0 - \alpha_1,\ e^{-2t} = \alpha_0 - 2\alpha_1$$
$$\Rightarrow \alpha_1 = e^{-t} - e^{-2t},\ \alpha_0 = 2e^{-t} - e^{-2t}$$

Hence $e^{At} = [\,\alpha_0\ \alpha_1\,;\,-2\alpha_1\ \alpha_0 - 3\alpha_1\,] = [\,2e^{-t} - e^{-2t}\ e^{-t} - e^{-2t}\,;\,-2e^{-t} + 2e^{-2t}\ -e^{-t} + 2e^{-2t}\,]$ — the same as the Laplace method ☒.

Problem 4 – Zero-input response

With $x(0) = [\,1\,;\,0\,]$ and the same A: $x(t) = \Phi(t)x(0) = [\,2e^{-t} - e^{-2t}\,;\,-2e^{-t} + 2e^{-2t}\,]$; both states decay to zero because the eigenvalues are negative.

Stability and the state model

- Asymptotically stable iff all eigenvalues of A (roots of $|sI - A| = 0$) have negative real parts.
- Eigenvalues are invariant under similarity transformation $x = Mz$; the transfer function is also invariant.
- State feedback $u = -Kx$ changes A to $(A - BK)$; if (A, B) is controllable the eigenvalues can be placed anywhere (pole placement) by suitable K.

Quick-revision formulas

$$x' = Ax + Bu\ ,\ y = Cx + Du \mid G(s) = C(sI - A)^{-1}B + D$$

$$\Phi(t) = e^{At} = L^{-1}[(sI - A)^{-1}] = M\, e^{\Lambda t}\, M^{-1}$$

$$x(t) = \Phi(t)x(0) + \int \Phi(t - \tau)Bu(\tau)d\tau$$

$$Q_c = [B\ AB \dots A^{n-1}B]\ ,\ Q_o = [C^T\ A^TC^T \dots] \rightarrow rank\ n$$

Important Questions

- *Define state, state variable, state model; obtain state model of a given electrical network / T.F. (5M/10M)*
- *Derive the state model from a block diagram. (5M)*
- *Diagonalise the given matrix A and find the modal matrix. (10M)*
- *Find the state transition matrix and solution x(t) for a given system by Laplace/Cayley–Hamilton method. (10M)*
- *List properties of STM with proof. (5M)*
- *State and test Kalman's controllability and observability criteria for the given system. (10M)*